Exotic terranes of western California

M. O. McWilliams

Geophysics Department, Stanford University, Stanford, California 94305, USA

D. G. Howell

US Geological Survey, Menlo Park, California 94025, USA

Numerous distinct geological terranes compose the North American Cordillera[1]; there may be as many as 50 terranes in California alone[2]. Critical to deciphering the history of Cordilleran tectonic assembly is an understanding of the displacement history of individual terranes. It is therefore important to know: (1) whether a terrane has undergone significant motion with respect to the stable craton (that is, whether it is allochthonous or exotic); (2) if so, when relative motion started and stopped; (3) from where an individual terrane originated; and (4) the nature of inter-terrane movements. We consider here the problem of determining whether the now-juxtaposed Salinian and Stanley Mountain terranes of California became amalgamated at or near their present position with respect to cratonic North America, or if they collided at a considerable distance from their present positions and were later accreted to North America as a composite package. The palaeomagnetic data that we present indicate that the latter was the case.

Terranes are fault-bounded geological entities of regional extent, each characterized by a geological history that is different from that of adjacent terranes. Such geological histories are determined from stratigraphical successions preserved within terranes, but in some cases this information is obscured owing to tectonic or sedimentological disruption or metamorphic overprinting. Composite terranes are produced by the amalgamation of two or more terranes. The southern Coast Ranges of California west of the San Andreas Fault consist of two composite tectonostratigraphical packages called the Salinian and Sur-Obispo terranes (Fig. 1). Basement rocks of the Salinian terrane are Middle and Upper Cretaceous granite plutons and older high-temperature metasedimentary rocks bounded on the west by the Sur-Nacimiento Fault. To the west lies the Sur-Obispo composite terrane, represented by the Stanley Mountain and San Simeon terranes. The Stanley Mountain terrane consists of Middle Jurassic ophiolite and an overlying thick sequence of Upper Jurassic to Upper Cretaceous forearc–basin strata. Structurally below the Stanley Mountain terrane is Franciscan mélange of the San Simeon terrane. The Stanley Mountain strata closely resemble, and are coeval with, the Great Valley sequence[3], which is typified by exposures along the west sides of the Sacramento and San Joaquin Valleys. Truncation by faults and unconformities disrupts the section in most places, but a nearly complete section of 4,000 m of Upper Jurassic and Cretaceous strata is exposed in the southeastern San Rafael Mountains[4]. These forearc–basin strata are demonstrably allochthonous with respect to Salinian basement, yet in late Cretaceous time the two terranes seemingly collided to form a new amalgamated terrane, as Campanian and younger sedimentary rocks of both terranes are petrographically identical and are characterized by non-marine horizons of locally derived granitic detritus[5,6] (Fig. 2). Several coherent packages of Upper Cretaceous arkosic strata in the San Simeon terrane may indicate continuity with the Salinian and Stanley Mountain terranes; however, equivocal sedimentary sequences which lap across these terranes are no older than early Tertiary.

Commencing in Campanian time with the paralic granite–grus lithofacies, the forearc took on a borderland-like configuration with detritus spilling into contrasting depositional settings on the newly juxtaposed terranes[6,7]. An episode of wrench faulting accompanied the suturing presumably along a now-distant margin of the Cordillera because Upper Cretaceous and Lower Tertiary strata contain numerous clasts of siliceous volcanic porphyries admixed with the granitic debris, yet such a volcanic provenance does not exist on either the Salinian or Stanley Mountain terranes.

Palaeomagnetic inclination data from Upper Cretaceous strata of the Salinian terrane indicate that deposition occurred as much as 2,500 km south of their present location (ref. 8; M. E. Williams, M. Schnapp and A. Cox, in preparation; D. E. Champion, personal communication). We report here new data which indicate a similar exotic origin for the Jurassic and Cretaceous rocks of the Stanley Mountain terrane.

To determine the palaeolatitude of the Stanley Mountain terrane in Jurassic time, samples of pillow basalt were collected at 10 sites spanning an 800 m section of ophiolite at Stanley Mountain (Figs 1, 2). Alternating field (a.f.) demagnetization of pilot specimens from each site yielded stable, well grouped magnetization directions at eight sites after removal of young secondary magnetizations of probable recent viscous origin; one example is shown in Fig. 3*a*. After partial demagnetization in peak alternating fields of 60–80 mT, the eight mean directions are well grouped and define an overall *in situ* mean direction $D = 053°$, $I = +54°$, $\alpha_{95} = 8°$. This direction is distinct from the present geomagnetic and axial dipole field directions for the sampling locality, as well as from any expected field direction for stable North America since early Mesozoic time[9]. A reasonable explanation for the *in situ* direction is that it represents a sampling of a primary Jurassic magnetization (TRM or high-temperature CRM) and should therefore be restored to palaeohorizontal attitude, thereby obtaining the palaeolatitude at which the rocks acquired their magnetization. If the magnetization is primary, the overall tectonically corrected mean direction is $D = 041°$, $I = -27°$, $\alpha_{95} = 8°$, corresponding to a palaeolatitude of $14 \pm 7°$. Structural attitudes of sedimentary rocks which immediately overlie the basalts were used as a palaeohorizontal reference indicator. If the age of the pillow basalt is 160 ± 5 Myr, as indicated by isotopic data from other Coast Range ophiolite samples[10], then a northerly or southerly

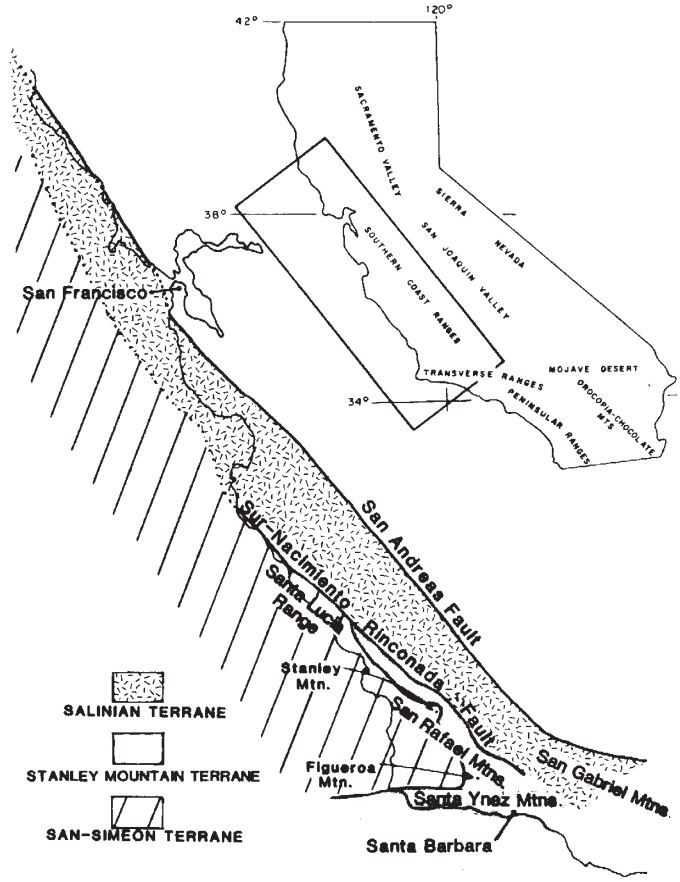


Fig. 1 Geographical sketch map of Stanley Mountain, San Simeon and Salinian terranes in western California.

0028-0836/82/200215—03$01.00

© 1982 Macmillan Journals Ltd

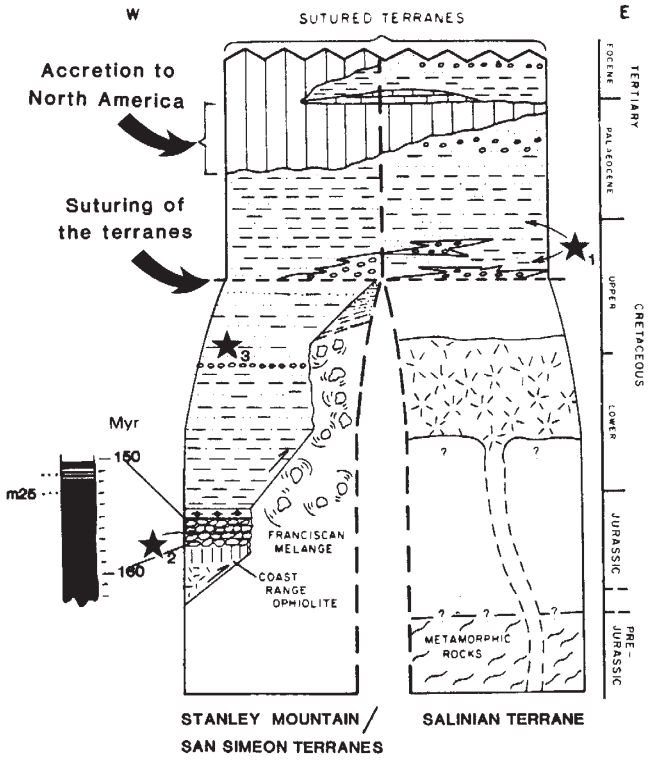


Fig. 2 Generalized stratigraphical columns for Salinian and Sur-Obispo composite terranes. Suturing of the two terranes was probably completed by the late Cretaceous, as equivalent strata lap onto both terranes after this time. ★, Stratigraphical positions of palaeomagnetic studies (*a*, D. E. Champion, personal communication; *b*, ref. 8 and M. E. Wilson, M. Schnapp and A. Cox, in preparation). The inset on the left is magnetic reversal time scale[11] (black = normal polarity).

palaeolatitude cannot be invoked, for the geomagnetic field was of mixed polarity at this time[11] (see reversal time scale inset, Fig. 2).

If the magnetization were of secondary origin and dated from Tertiary time, an exact tectonic correction would not be possible because the structural attitude at the time of remagnetization would be unknown. One possibility would be that the basalts acquired their magnetization after tectonic tilting and were subsequently rotated 50–60° clockwise about a vertical axis. At the locality where the samples were collected, the basalts form part of a steeply dipping homoclinal sequence, so that a fold test was not possible. There is no evidence, however, for Tertiary remagnetization by a thermal mechanism, for isotopic ages are not reset and reheating effects are not visible in hand specimen or thin section.

The sedimentary record indicates that the Stanley Mountain and Salinian terranes probably collided in the late Cretaceous. To determine the palaeolatitude of the Stanley Mountain terrane at approximately the time of collision, samples of Upper Cretaceous (Cenomanian–Santonian) flysch strata were collected at 10 sites in the San Rafael Mountains near Figueroa Mountain (Figs 1, 2). After a.f. demagnetization in peak fields up to 80 mT, a single stable component of magnetization was isolated (an example is shown in Fig. 3*b*). The overall *in situ* mean direction from these 10 sites is $D = 002°$, $I = +74°$, $\alpha_{95} = 10°$, which is significantly different from both the present geomagnetic field direction and the axial dipole field direction at the sampling locality, suggesting that it is not of recent origin. Also, this overall mean direction is significantly different from any geomagnetic field direction since early Mesozoic time derived from stable North America[9]. On tectonic correction to palaeohorizontal, the overall mean direction is $D = 023°$, $I = +12°$, $\alpha_{95} = 11°$. The minor decrease in overall precision accompanying this correction is not statistically significant and may be due to pre-magnetization slumping as noted in other palaeomagnetic studies of turbidites[8]. The slightly skewed distribution of site mean directions may be due to our inability to make exact tectonic corrections which compensate for this post-depositional slumping.

If the magnetization of these samples is primary and dates from the time of deposition or soon thereafter, the tectonically corrected inclination of $12 \pm 10°$ would correspond to a palaeolatitude of $6 \pm 5°$ (N or S). This range is slightly lower than the range of apparent palaeolatitudes from the Pigeon Point Formation on the Salinian terrane (ref. 8, and M. E. Wilson, M. Schnapp and A. Cox, in preparation), but if a Northern Hemisphere interpretation is invoked (Fig. 4), this might be expected, as the Pigeon Point strata are slightly younger (Campanian–Maestrictian). An alternative explanation for the palaeolatitude discrepancy could be that the suturing of the two terranes might not have been completed by this time.

The palaeomagnetic data shows that (1) the Salinian composite terrane lay in low ($21 \pm 7°$) palaeolatitudes[8] in Campanian–Maestrictian time, and occupied higher ($25 \pm 3°$) palaeolatitudes[10] in early Palaeocene time; (2) the Stanley Mountain terrane lay in slightly lower ($6° \pm 5°$) palaeolatitudes in Cenomanian–Santonian time, and occupied slightly higher ($14° \pm 7°$) palaeolatitudes at about 160 ± 5 Myr; either a north or south palaeolatitude is possible; (3) a simple displacement model (solid line in Fig. 4) would place the Sur-Obispo composite terrane at 14° S at 160 Myr, at 6° N at about 90 Myr and at 21° N at about 70 Myr, after colliding with the Salinian composite terrane; and (4) alternatively, the Stanley Mountain terrane could have occupied low northerly palaeolatitudes from Jurassic to Cretaceous time, beginning its major northward migration at about the time it collided with the Salinian terrane (dashed line in Fig. 4). Coupled with the geological arguments, these data indicate that the late Cretaceous collision of the two terranes occurred in low palaeolatitudes, far from the present California continental margin.

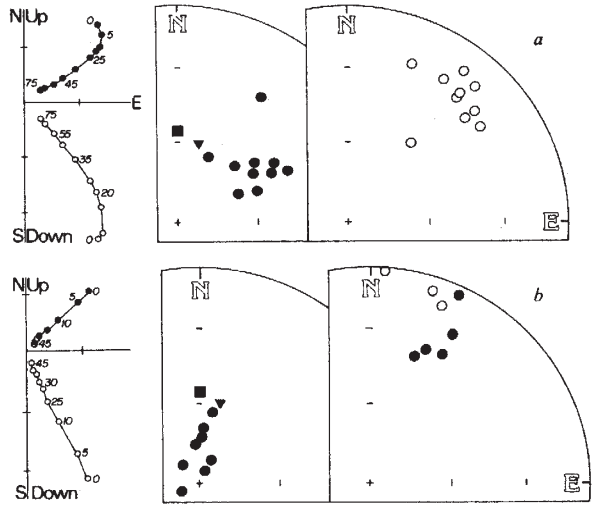


Fig. 3 *a*, Palaeomagnetic results from Stanley Mountain pillow basalts. Left: orthogonal vector projection of a.f. demagnetization experiment; circles represent projection of magnetization vector on the horizontal (●) or east–west vertical (○) plane; axis ticks 0.2×10^{-3} A m^{-1}. Numbers are peak a.f. treatment in units of 10^{-3} T. Centre: equal area projection of site mean magnetization directions after a.f. demagnetization. No tectonic correction is applied; each direction plotted is the average of six individually oriented samples. Solid symbols represent positive (downward) inclinations. ■ and ▼ are axial dipole field and present geomagnetic field directions respectively. Right: as in centre, but with magnetization directions restored to palaeohorizontal. Open symbols represent negative (upward) inclination. *b*, Palaeomagnetic results, Upper Cretaceous sedimentary rocks from Figueroa Mountain. Left: as above, axis ticks 0.1×10^{-3} A m^{-1}. Centre: site mean directions before tectonic correction; conventions as above. Right: site mean directions before tectonic correction; conventions as above.

©1982 Nature Publishing Group

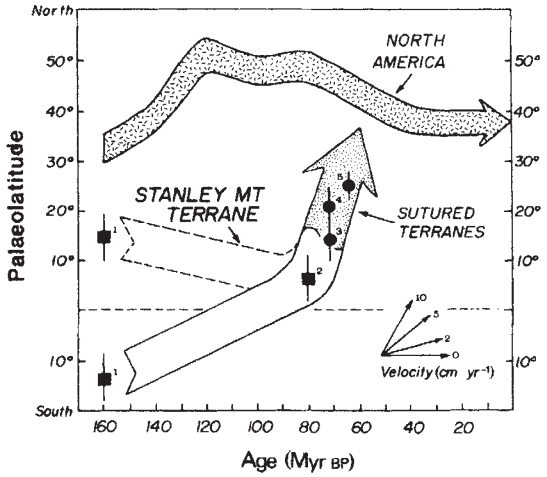


Fig. 4 Palaeolatitude of the Salinian and Stanley Mountain terranes as a function of time, compared with palaeolatitude of 'target area' on North America for present-day position, 35–39° north latitude. Stippled path represents the two terranes after suturing. The North American and Salinian/Stanley Mountain tracks intersect at ~50–55 Myr BP to allow for Eocene reconstruction of the Butano-Point of Rocks submarine fan system. ■, Data from Sur-Obispo terrane; ●, data from Salinian terrane. (1) and (2) are from this study; (4) from ref. 8; (3) from M. E. Wilson, M. Schnapp and A. Cox, in preparation; and (5) D. E. Champion, personal communication). Arrows in lower right give northward velocity in cm yr^{-1} for reference. The dashed and solid lines represent alternative palaeolatitudes for the Stanley Mountain terrane depending on the choice of magnetic polarity.

The classic sequence of events proposed for the evolution of the western Cordillera involves three general episodes[12–14]: (1) a late Jurassic to early Tertiary subduction regime with major deposition in an arc–trench gap configuration; (2) early Tertiary proto-San Andreas faulting and a borderland setting during an early to middle Tertiary episode of oblique subduction along a flattened Benioff–Wadati zone; and (3) continental margin transform faulting resulting in tectonic splintering and additional borderland development. If recent palaeomagnetic studies (ref. 8; M. E. Wilson, M. Schnapp and A. Cox, in preparation; D. E. Champion, personal communication), including our own, are accurate and basin analyses[4,6,7] correct, both the timing of events and nature of origin of the continental margin must be reinterpreted.

Analysis of the 1,500 m thick fluvio-deltaic sequence in the southeastern Stanley Mountain terrane indicates that the Stanley Mountain and Salinian terranes were probably attached by ~72 Myr BP. Thus the Stanley Mountain and Salinian terranes, both of which come from quite different parent terranes, must have travelled northwards together, despite the remarkable parallelism of Cretaceous depositional environments between the Stanley Mountain terrane and the Great Valley sequence of the Sacramento and San Joaquin Valleys. However, the Stanley Mountain/Salinian composite package was probably emplaced in California by the early Eocene, based on the inference that submarine fans sequences of this age[15] lap across these terranes and other terranes autochthonous to California.

The palaeomagnetic data presented here and elsewhere indicate a large northward component of relative plate motion along the western margin of North America during the Cretaceous and early Tertiary. Several late Mesozoic terranes presumably originated at the latitude of present-day Central America and moved northwards to the California area by late Palaeocene time. Plate motion reconstructions (ref. 16 and D. C. Engebretson, in preparation) demonstrate that Farallon–North America relative motion for most of the Cretaceous would have been largely orthogonal to a north-west trending margin of western North America. Kula plate motion relative to North America had a significant northward component at high latitudes, but owing to an Euler pole position in northwestern South America, Kula–North America relative motion in the Central American regions would also have been largely orthogonal (ref. 17 and D. C. Engebretson, in preparation).

Two possible ways to account for the northward shifting of terranes are immediately apparent[17]: (1) reconstruct the late Mesozoic Pacific basin plates including now vanished plate(s) whose configurations would account for the northward slip required by the palaeomagnetic data and for the episodic plutonism mapped all along the Cordillera, and/or (2) palinspastically restore Central America and the ancient Caribbean plate westwards to increase the distance from the Kula–North America Euler pole in late Mesozoic time, so that Kula plate motion could better account for the northward component of motion of the Stanley Mountain and Salinian terranes.

We thank Anne Demitrack, Brian Hausback, Bill Keller, Peter Lellis and Norman Sleep for assistance in sample collection, David Engebretson and Ben Page for editorial comments, and especially Michel Debiche and Anne Demitrack for many laboratory measurements. The research was supported by grant PRF-11405 from the Petroleum Research Fund of the American Chemical Society.

Received 21 January; accepted 9 March 1982.

1. Coney, P. J., Jones, D. L. & Monger, J. W. H. *Nature* **288**, 329–331 (1980).
2. Blake, M. C., Howell, D. G. & Jones, D. L. *U.S. geol. Surv. Open File Map Ser.* (in the press).
3. Page, B. M., in *The Geotectonic Development of California* (ed. Ernst, W. G.) (Prentice-Hall, Englewood Cliffs, New Jersey, 1981).
4. Howell, D. G. *et al.* in *Soc. Economic Paleontologists and Mineralogists, (Pacific Sec.), Pacific Coast Paleogeogr. Field Guide* Vol. 2 (eds Howell, D. G., Vedder, J. H. & McDougall, K.) 1–46 (1977).
5. Howell, D. G., McLean, H. & Vedder, J. H. *EOS* **61**, 949 (1980).
6. Vedder, J. H., Howell, D. G. & McLean, H. *Mem. Am. Ass. petrol. Geol.* (in the press).
7. Howell, D. G. & Vedder, J. H. *Soc. Economic Paleontologists and Mineralogists, (Pacific Sec.), Pacific Coast Paleogeography Symp.* Vol. 2 (eds Howell, D. G. & McDougall, K. A.) 107–116 (1978).
8. Champion, D. E., Gromme, C. S. & Howell, D. G. *Soc. Economic Paleontologists and Mineralogists, (Pacific Sec.), Field Trip Guide, Upper Cretaceous and Paleocene Turbidites, Central California Coast*, 53–56 (1981).
9. Irving, E. *Can. J. Earth Sci.* **16**, 669–694 (1979).
10. Hopson, C. A., Mattinson, J. M. & Pessagno, E. A. in *The Geotectonic Evolution of California* (ed. Ernst, W. G.) (Prentice-Hall, Englewood Cliffs, New Jersey, 1981).
11. Cox, A. in *The Geologic Timescale* (ed. Harland, W. B.) (Cambridge University Press, in the press).
12. Dickinson, W. R. *Rev. geophys. Space Phys.* **8**, 813–860 (1970).
13. Dickinson, W. R. *Am. J. Sci.* **272**, 551–576 (1972).
14. Dickinson, W. R. & Rich, E. I. *Bull. geol. Soc. Am.* **83**, 3007–3024 (1972).
15. Nilsen, T. H. & Clarke, S. H. *U.S. geol. Surv. Professional Pap.* **925**, 1–64 (1975).
16. Morgan, W. J. in *The Sea* (ed. Emiliani, C.) (Wiley, New York, 1980).
17. Page, B. M. *Am. J. Sci.* (in the press).

Implications of Palaeozoic phosphorites in the northern Sierra Nevada range

Robert J. Varga

Union Oil Research Center, PO Box 76, Brea, California 92621, USA

Marine phosphorites are known to occur within multiply deformed Ordovician/Silurian and Upper Devonian rocks of the northern Sierra Nevada range, California. Although the phosphatic sediments constitute a small percentage of total lithologies in the area, they provide unique information regarding the age and palaeogeographical setting of these rocks. The mode of occurrence of the Sierran phosphorites and their association with black shale and chert suggest that they may have formed along a continental margin which experienced strong equatorial or trade wind belt upwelling. Consideration of worldwide plate reconstructions for the early to middle Palaeozoic shows that the present western margin of North America was one of the few favourable sites for phosphate formation during this time period. Thus, the presence of marine phosphorites within Palaeozoic rocks of the northern Sierra Nevada is consistent with arguments which suggest that these rocks were deposited near the western margin of North America. Although considerable margin-parallel translations cannot be ruled out for these rocks, they do not appear to constitute a far-travelled microplate derived from a distant, trans-oceanic site.

0028–0836/82/200217—04$01.00
© 1982 Macmillan Journals Ltd